In[103]:=

```
coords = {t, r, θ, ϕ};

(*Defining the metric*)
g = {{f[r], 0, 0, 0}, {0, h[r], 0, 0}, {0, 0, r^2, 0}, {0, 0, 0, r^2 Sin[θ]^2}};

(*Defining the inverse metric*)
ginv = Simplify[Inverse[g]];

n = Length[coords];
Gammachris[ρ_, μ_, ν_] :=
 1/2 Sum[ginv[[ρ, σ]] (D[g[[σ, ν]], coords[[μ]]] + D[g[[σ, μ]], coords[[ν]]] - D[g[[μ, ν]], coords[[σ]]]), {σ, 1, n}]

RieTensor[ρ_, σ_, μ_, ν_] :=
 D[Gammachris[ρ, ν, σ], coords[[μ]]] - D[Gammachris[ρ, μ, σ], coords[[ν]]] +
  Sum[Gammachris[ρ, μ, λ] × Gammachris[λ, ν, σ], {λ, 1, n}] -
  Sum[Gammachris[ρ, ν, λ] × Gammachris[λ, μ, σ], {λ, 1, n}]
RiccTensor[μ_, ν_] := Sum[RieTensor[ρ, μ, ρ, ν], {ρ, 1, n}]

Print["Non-zero components of Ricci tensor R_{μν}:"];

Do[Module[{val = Simplify[RiccTensor[μ, ν]]},
   If[val =!= 0, Print["R_", coords[[μ]], coords[[ν]], " = ", val]]], {μ, 1, n}, {ν, 1, n}];
```

Non-zero components of Ricci tensor R_{μν}:

$$\text{R\_tt} = \frac{r\,h[r]\,f'[r]^2 + f[r]\left(r\,f'[r]\,h'[r] - 2\,h[r]\left(2\,f'[r] + r\,f''[r]\right)\right)}{4\,r\,f[r]\,h[r]^2}$$

$$\text{R\_rr} = \frac{f[r]\left(4\,f[r] + r\,f'[r]\right)h'[r] + r\,h[r]\left(f'[r]^2 - 2\,f[r]\,f''[r]\right)}{4\,r\,f[r]^2\,h[r]}$$

$$\text{R\_}\theta\theta = \frac{1}{2}\left(2 - \frac{2 + \frac{r\,f'[r]}{f[r]}}{h[r]} + \frac{r\,h'[r]}{h[r]^2}\right)$$

$$\text{R\_}\phi\phi = \frac{\text{Sin}[\theta]^2\left(-r\,h[r]\,f'[r] + f[r]\left(-2\,h[r] + 2\,h[r]^2 + r\,h'[r]\right)\right)}{2\,f[r]\,h[r]^2}$$